Modeling light pollution using geospatial, meteorologoical, energy usage, and population data

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1 Executive Summary

Problem Statement Light pollution is a night-time phenomenon caused by the atmospheric scattering of excess light from things such as street lamps or other outdoor light fixtures that then bounces off of molecules in the air. Light pollution is commonly measured as night sky brightness (in magnitudes per square arcsecond), but is typically only directly measured at sites like observatories. Even then, data from these observatories is rarely accessible in real-time. The Globe at Night campaign created a monitoring network for observatories across the world to provide night sky brightness data for public use, with data going back as far as 2014. Light pollution, being an atmospheric phenomenon, is heavily impacted by weather conditions. The objective for this project is to predict light pollution (night sky brightness) at a given location using primarily meteorological, energy usage, and population data.

Major Concerns/Assumptions TO-DO

Summary of Findings TO-DO

Recommendations TO-DO

2 Problem Background

2.1 Problem Description

The use of artificial light at night (ALAN) in outdoor spaces—e.g., through the use of light fixtures such as street lamps or flood lights—causes what is known as light pollution. More specifically, improper and excessive use of ALAN, which has risen dramatically alongside rapid urbanization and industrialization around the globe, is the primary source of light pollution. The definition of light pollution varies across different agencies, but the common denominator amongst them all is that it is the culmination of the unwanted, adverse aspects of ALAN. There are three main components that make up light pollution: glare, light trespass, and skyglow; for the purposes of this project, however, the focus is solely on skyglow[7]. Skyglow, the most broadly impactful aspect of light pollution, is the result of light being emitted upwards and then bouncing off of particles in the Earth's lower atmosphere, scattering. It obscures the natural night sky, preventing those in the city from being able to marvel at the stars. Those with exceptional vision and access to pristine night skies are able to see up to 7,000 stars with the naked eye, however, that number plummets to fewer than 100 for the same observer in an urban area[5].

As it currently stands, light pollution-free skies are becoming more and more difficult to find. Over 99% of US and European populations are affected by some level of light pollution, and a recent analysis found that skyglow is rapidly worsening at a rate of nearly 10% per year[4]. Modeling the skyglow component of light pollution, typically referred to as night sky brightness, is a crucial aspect of many light pollution studies. Unfortunately, there is no definitive way to assess skyglow, and each method has its own pros and cons. Light pollution data for modeling often comes from a combination of ground-based measurements and satellite data. Ground-based measurements are most often taken using a sky quality meter (SQM), but have also been derived from citizen-science reports of naked-eye limiting magnitude (NELM). NELM is the magnitude (a logarithmic scale describing the apparent brightness of a star) of the faintest visible star to a naked-eye observer; the lower the magnitude, the brighter the object. The SQM is a handheld sensor that measures the brightness of the sky directly overhead in units of magnitudes per square arcsecond.

Due to the complexity of light emission behavior and limitations in computing power, research into different modeling methods is still ongoing [3]. Most light pollution models are numerical models based on a series of functions that describe upward light flux and atmospheric scattering. Many more radiative transfer-based numerical models exist, but are often tailored towards modeling meteorological conditions and therefore not suitable for modeling light pollution specifically [3]. Recently, it was found that most, if not all light pollution models are significantly, systematically biased through implicitly assuming that all particles in the lower atmosphere are homogenous and spherical (Mie theory), despite being a safe assumption for daytime light models [2]. A popular global map known as the New world atlas of artificial night sky brightness [1] is another numerical model utilizing both ground-based and satellite measurements, but assumed a perfectly clear, calm, atmosphere over the entire globe. This limitation, albeit understandable, greatly impacts the accuracy of its output. Atmospheric conditions vary widely across regions. Not only that, but atmospheric conditions are always changing as well. Consecutive measurements of skyglow in the exact same location, even just hours apart, could produce different results.

While acknowledging that the extent of light pollution is influenced by a multivariate set of economic indicators [6], the current model specification is limited to assessing the predictive power of population metrics and energy consumption data.

2.2 Data Description

Night-Sky Brightness Globe at Night (GaN) The GaN-MN project, an extension of the original Globe at Night project, is a global night sky brightness monitoring network using a commercially available meter (SQM-LE by Unihedron) for long-term monitoring of the light pollution conditions in different places around the world. https://globeatnight.org/gan-mn/

Meteorological Data ERA5 provides hourly estimates for a large number of atmospheric, ocean-wave and land-surface quantities. An uncertainty estimate is sampled by an underlying 10-member ensemble at three-hourly intervals. Ensemble mean and spread have been pre-computed for convenience. Such uncertainty estimates are closely related to the information content of the available observing system which has evolved considerably over time. They also indicate flow-dependent sensitive areas. To facilitate many climate applications, monthly-mean averages have been pre-calculated too, though monthly means are not available for the ensemble mean and spread. https://cds.climate.copernicus.eu/datasets/reanalysis-era5-pressure-levels?tab=overview

Zip-Code Populations A tabulation of the total population per zip code per the 2020 U.S. Census. https://data.census.gov/table

Electric Retail Service Territories A table of electric power retail service territories in the U.S. These are areas serviced by electric power utilities responsible for the retail sale of electric power to local customers, whether residential, industrial, or commercial. https://ornl.opendatasoft.com/explore/dataset/electric-retail-service-territories/information/

Population-Weighted Centroids for Zip Code Tabulation Areas (ZCTAs) This dataset denotes ZIP Code centroid locations weighted by population. https://hudgis-hud.opendata.arcgis.com/datasets/d032efff520b4bf0aa620a54a477c70e/about

2.3 Exploratory Data Analysis

Night-Sky Brightness The Globe at Night dataset only contains thirteen sites and night sky brightness readings are relatively sparse (63% missing). Additionally the minimum time resolution of measurements is to the UTC hour instead of the local hour.

Meteorological Data There sure is a lot of ERA5 data.

Electric-Service Provider Coverage The service territories dataset is missing zip-code data for approximately 58% of electrical service providers.

Population 4320 zip-codes are missing between the intersection of the centroids and population datasets. This is due to zip-codes that did not have population data in the 2020 census data.

3 Methodology

3.1 Feature Selection

Night-Sky Brightness Missing brightness readings are dropped from the training data entirely. The time of observation is assumed to be close enough to local time, so no time zone adjustments are made. The brightness readings are assumed to be correlated to the time as a cyclical feature, so the month and hour of observation are encoded as wave functions at an angular offset from typical brightness zeniths.

Meteorological Data TO-DO

Electric-Service Provider Rows with missing zip-code data for service providers are dropped from the training data due to the inability to map given providers to sensible zip-codes.

Population The zip-code centroids with missing population data are dropped from the training data.

Expected Energy Usage The assumption is made that energy usage by nearby population centers is correlated to their nearest electric-service providers. Using the population-centroids of each zip-code, the nearest electric-service provider is associated with each population center. The nominal energy usage of each population center is then encoded as a sine function of the month of observation to differentiate between the seasonal energy usage patterns. The `SUMMR_PEAK` and `WINTR_PEAK` values are used as the amplitudes of the wave functions for the respective seasons.

Distance from Population Centers Using the Haversine formula, the great circle distance between the nearest zip-code population centroid and the observation site is calculated and included as a feature. Since the radiance from light-sources are known to decay with distance, this distance is encoded as its inverse square. Since this predictor is known to have a direct physical relationship with the target variable, it is not mean-centered or scaled.

3.2 Model Choice

TO-DO. Probably MARS? Model-stacking with linear regression, SVM, GAM, GLM, and whatever else is within computational achievability.

3.3 Validation Approach

Unless significant computational throughput problems are encountered, 10-fold cross-validation will be used for model training and evaluation.

4 Results

4.1 Model Results and Performance

TO-DO

4.2 Key Findings

TO-DO

5 Conclusion

Summary TO-DO

Key Insights TO-DO

Final Recommendations and Impact TO-DO

References

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